

Smart Traffic Lights Using Local AI for Emergency Vehicle Route Coordination

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Abstract—Smart traffic-light systems can use information from the road to respond to changing traffic conditions. Traditional fixed-time traffic lights are simple and inexpensive, but they may not respond quickly when traffic conditions change. This paper presents a conceptual local-AI architecture in which street cameras provide traffic information to smart-pole computers for local processing. Selected information may also be forwarded to a cloud server for storage and wider analysis. The main research focus is emergency vehicle coordination, particularly the detection of an ambulance and the coordination of nearby traffic signals to create a clear route. The proposed architecture is described through a system workflow, comparison of traffic-control approaches, and a conceptual emergency-vehicle decision process. The numerical comparison values included in the source material are treated as illustrative rather than experimentally validated results.

Index Terms—Smart Traffic Lights, Local AI, Traffic Signal Control, Machine Vision, Emergency Vehicle Detection, Edge Computing

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timing plans. They are simple and inexpensive, but fixed timing may not respond well when traffic conditions change suddenly.

Recent approaches use cameras and artificial intelligence (AI) to estimate traffic conditions and adapt traffic-signal operation. Research on traffic-signal control has investigated AI-based methods, including deep reinforcement learning, for adapting signals to changing traffic conditions [1]. Machine vision has also been studied for adaptive traffic-light control using camera-based traffic information [2].

Cloud-based AI systems can send camera information to a central server for processing. This can provide access to greater computational resources, but it introduces dependence on network connectivity and may add communication delay.

Local smart-camera systems move part of the processing closer to the road. Such an architecture may reduce the amount of information transmitted over the network and support faster local decisions.

This paper focuses on a conceptual local-AI traffic-light system with an emergency-vehicle coordination capability. The objective is to detect an emergency vehicle and coordinate nearby traffic signals so that a clear route can be created.

II. LITERATURE REVIEW

Rasheed et al. reviewed deep reinforcement learning approaches for traffic signal control and discussed their use for adaptive traffic management [1]. Their review provides a broad overview of learning-based traffic-signal control and associated challenges.

Gorodokin et al. investigated optimization of adaptive traffic-light control modes using machine vision [2]. Their work demonstrates how camera-based traffic information can be used as an input for adaptive signal operation.

Emergency-vehicle-centered traffic signal control has also been studied. Cao et al. proposed an emergency-vehicle-centered signal control approach designed to support emergency vehicles while reducing the negative effect on conflicting traffic directions [3]. This work is directly relevant to emergency vehicle priority at congested intersections.

These studies support the use of AI, machine vision, and emergency-vehicle signal control as tools for intelligent traffic management. The present paper narrows the problem to local processing and emergency-vehicle route coordination.

III. PROBLEM STATEMENT AND RESEARCH GAP

Existing traffic systems can operate effectively under regular traffic conditions, particularly when traffic patterns are predictable. However, an emergency situation such as an ambulance approaching a congested intersection requires rapid coordination between nearby traffic signals.

The research problem considered in this paper is:

How can a local-AI traffic-light system detect an emergency vehicle and coordinate nearby signals to create a clear route while continuing to manage normal traffic conditions?

The research gap is therefore the integration of local traffic analysis with emergency-vehicle route coordination in a single traffic-signal architecture.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed architecture follows four main stages.

- 1) **Street Cameras:** Capture traffic conditions.
- 2) **Smart Pole Computers:** Process camera input locally and make a traffic decision.
- 3) **City Cloud Server:** Receive selected information for storage and wider analysis when required.

- 4) **Traffic Lights:** Receive the control decision and change their signal states.

The basic information flow is:

Street Cameras → Smart Pole Computers → Traffic Lights. (1)

Selected information can additionally be transmitted to the cloud:

Smart Pole Computers → City Cloud Server. (2)

This separates time-sensitive local control from optional cloud-based storage and analysis.

V. SYSTEM FLOWCHART

Fig. 1 shows the conceptual processing flow of the proposed local-AI system.

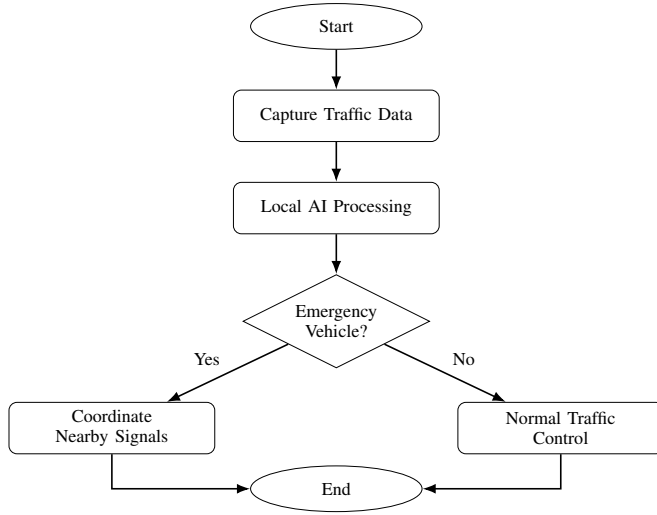


Fig. 1. Conceptual flowchart of the proposed local-AI system

VI. SYSTEM COMPARISON

The source material compares three traffic-light approaches using decision time, internet usage, and setup cost. These values are retained below, but they are explicitly treated as illustrative values because the supplied material does not provide experimental methods or measurements supporting them.

TABLE I
ILLUSTRATIVE COMPARISON OF TRAFFIC-LIGHT APPROACHES

System	Decision Time	Internet Use	Setup Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

The values in Table I should not be interpreted as measured performance. A real comparison would require the same hardware, traffic conditions, datasets, and evaluation procedure for all three approaches.

VII. EMERGENCY VEHICLE COORDINATION

The proposed system adds an emergency-vehicle decision to the normal traffic-control process. When an emergency vehicle is detected, the local processor can identify the required direction of travel and communicate with nearby traffic-light controllers.

Emergency-vehicle signal preemption has been investigated in previous intelligent transportation research. Cao et al. studied emergency-vehicle-centered traffic signal control for supporting fast and safe passage through congested intersections [3].

A simplified emergency decision can be represented as

$$E = \begin{cases} 1, & \text{if an emergency vehicle is detected,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

When $E = 1$, the system enters emergency coordination mode. When $E = 0$, normal traffic control continues.

For a practical implementation, emergency-vehicle detection would need to be validated using a suitable camera dataset and measured using metrics such as precision, recall, false-positive rate, and detection latency.

VIII. MATHEMATICAL MODEL

For a local-AI architecture, total decision latency can be conceptually expressed as

$$T_d = T_c + T_p + T_s, \quad (4)$$

where T_c is camera capture and input-processing time, T_p is local AI processing time, and T_s is traffic-signal control time. A smaller T_d is desirable for time-sensitive applications, although the actual value must be determined experimentally.

Traffic density can be represented in a simplified form as

$$D = \frac{N_v}{L}, \quad (5)$$

where N_v is the number of detected vehicles and L is the selected road-section length.

IX. ALGORITHM

The conceptual emergency-vehicle coordination algorithm is shown in Algorithm 1.

X. IMPLEMENTATION CONSIDERATIONS

A practical implementation would require traffic cameras, local computing hardware, traffic-light controllers, and a communication mechanism between the local processor and signal controller.

The local processor would perform time-sensitive analysis, whereas the cloud could be used for storage, historical analysis, and system-wide monitoring. The separation can reduce the need to send every camera frame to a remote server.

The implementation should be evaluated using controlled experiments covering different traffic densities, lighting conditions, weather conditions, and emergency-vehicle approaches.

Algorithm 1 Local Emergency Vehicle Coordination

```
1: Capture camera input.
2: Process the input using the local AI model.
3: Detect vehicles and estimate traffic conditions.
4: if emergency vehicle is detected then
5:   Determine the emergency route.
6:   Coordinate nearby traffic signals.
7: else
8:   Continue normal traffic control.
9: end if
10: Record selected system information.
11: Return updated traffic-signal state.
```

XI. DISCUSSION

The reviewed literature supports the use of AI-based traffic signal control and machine vision for adaptive traffic management [1], [2].

Emergency-vehicle signal control has also been investigated as a method for reducing delays at congested intersections [3].

The proposed architecture extends these approaches by placing time-sensitive processing near the road and explicitly considering emergency-vehicle coordination.

Local processing may reduce dependence on continuous cloud communication for immediate decisions. However, this is an architectural expectation rather than an experimentally demonstrated result in the present paper.

The actual effectiveness of the proposed system would depend on emergency-vehicle detection accuracy, processing latency, communication reliability, traffic-light controller integration, and the ability to coordinate multiple intersections.

XII. LIMITATIONS

The present paper describes a conceptual architecture rather than a validated field deployment.

The supplied assignment material does not provide an experimental dataset, prototype measurements, or controlled tests for the proposed emergency-vehicle coordination system.

Camera-based detection can also be affected by lighting, occlusion, weather, camera position, and image quality. In addition, coordination with existing traffic infrastructure requires appropriate controller interfaces and safety procedures.

Therefore, the proposed architecture should be experimentally validated before claims about real-world latency, cost savings, or traffic improvement are made.

XIII. FUTURE WORK

Future work should include development of a working prototype and evaluation using real or simulated traffic data.

Important evaluation measures should include:

- emergency-vehicle detection accuracy;
- false-positive and false-negative rates;
- local processing latency;
- signal coordination time;
- route-clearing time;
- network bandwidth usage; and

- traffic-flow performance before and after coordination.

Further research can investigate edge-AI models, multi-intersection coordination, emergency-vehicle localization, and communication between adjacent smart traffic-light controllers.

XIV. CONCLUSION

This paper presented a conceptual smart traffic-light system using local AI for traffic analysis and emergency-vehicle route coordination. The architecture consists of street cameras, smart-pole computers, an optional city cloud server, and traffic-light controllers.

Previous research has investigated AI-based traffic-signal control and machine-vision-based adaptive traffic lights [1], [2]. Emergency-vehicle-centered traffic signal control has also been studied for improving emergency-vehicle passage through congested intersections [3].

Building on these areas, the proposed system focuses on a specific emergency-response problem: detecting an emergency vehicle and coordinating nearby signals to create a clear route.

The numerical comparison provided in the source material has been retained as illustrative information rather than experimental evidence. Future prototype development and controlled evaluation are required to determine the actual performance of the proposed system.

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